

CS-218 DATA STRUCTURES AND ALGORITHMS

LECTURE 25

BY UROOJ AINUDDIN



GRAPHS

BOOK 4 CHAPTER 8



IN THE LAST LECTURE...

We were introduced
to graph
terminology.

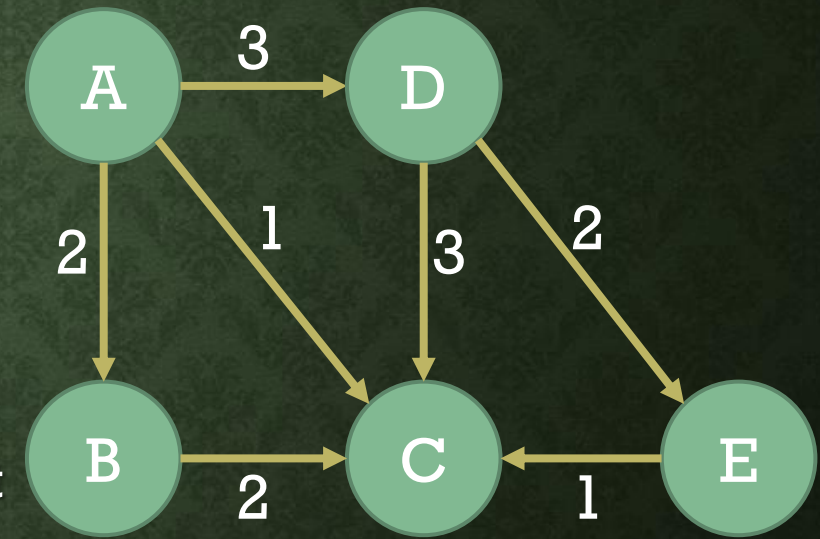
We stored the
graph in
memory.

We computed
path matrix
for the graph.

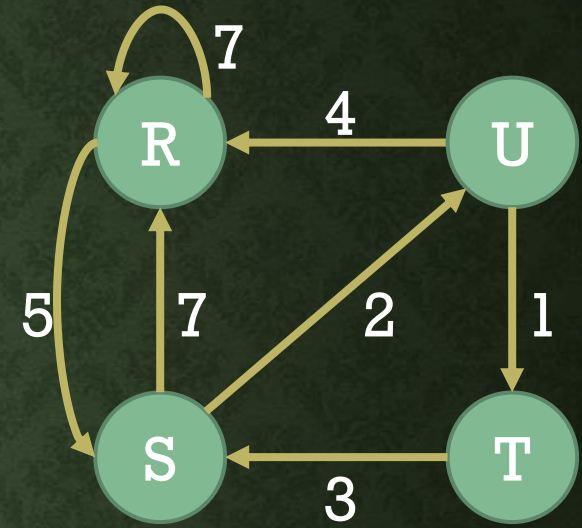


SHORTEST PATH ALGORITHM (SPA)

- The shortest path algorithm, also called Warshall's algorithm, is used to find the path with the shortest weight between each pair of nodes in a weighted graph.
- A **weighted graph** is a graph in which each edge is labelled with a number called its **weight**.
- The weight may represent any quantity of interest, like cost of using this edge, the time delay occurring on this edge, the distance covered in going from one end of this edge to the other, etc.



EXECUTION OF SPA



- The weight matrix W of this digraph is:

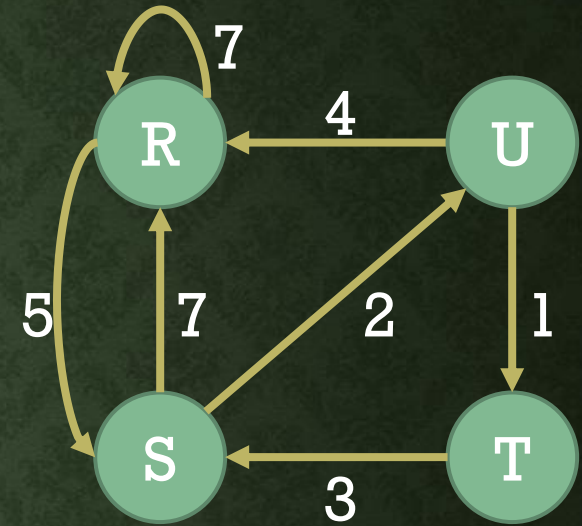
$$W = \begin{matrix} & \begin{matrix} R & S & T & U \end{matrix} \\ \begin{matrix} R \\ S \\ T \\ U \end{matrix} & \begin{bmatrix} 7 & 5 & 0 & 0 \\ 7 & 0 & 0 & 2 \\ 0 & 3 & 0 & 0 \\ 4 & 0 & 1 & 0 \end{bmatrix} \end{matrix}$$

- We construct $n+1$ matrices from Q_0 to Q_n , where n is the number of nodes in the graph.
- Q_0 is derived from W , replacing all zeros with ∞ to depict the absence of an edge between two nodes.

$$Q_0 = \begin{matrix} & \begin{matrix} R & S & T & U \end{matrix} \\ \begin{matrix} R \\ S \\ T \\ U \end{matrix} & \begin{bmatrix} 7 & 5 & \infty & \infty \\ 7 & \infty & \infty & 2 \\ \infty & 3 & \infty & \infty \\ 4 & \infty & 1 & \infty \end{bmatrix} \begin{bmatrix} RR & RS & - & - \\ SR & - & - & SU \\ - & TS & - & - \\ UR & - & UT & - \end{bmatrix} \end{matrix}$$



EXECUTION OF SPA



$$Q_0 = \begin{matrix} & \begin{matrix} R & S & T & U \end{matrix} \\ \begin{matrix} R \\ S \\ T \\ U \end{matrix} & \begin{bmatrix} 7 & 5 & \infty & \infty \\ 7 & \infty & \infty & 2 \\ \infty & 3 & \infty & \infty \\ 4 & \infty & 1 & \infty \end{bmatrix} \end{matrix} \begin{bmatrix} RR & RS & - & - \\ SR & - & - & SU \\ - & TS & - & - \\ UR & - & UT & - \end{bmatrix}$$

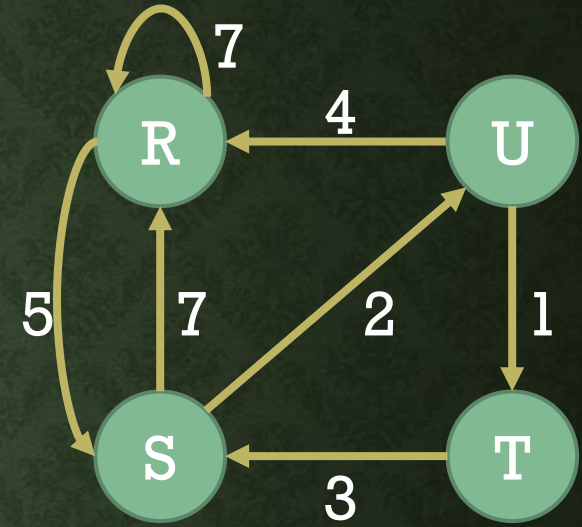
- Fix the first row and the first column to compute Q_1 .

$$Q_1 = \begin{matrix} & \begin{matrix} R & S & T & U \end{matrix} \\ \begin{matrix} R \\ S \\ T \\ U \end{matrix} & \begin{bmatrix} 7 & 5 & \infty & \infty \\ 7 & \blacksquare & & \\ \infty & & & \\ 4 & & & \end{bmatrix} \end{matrix} \begin{bmatrix} RR & RS & - & - \\ SR & \blacksquare & & \\ - & & & \\ UR & & & \end{bmatrix}$$

- Concentrate on the SS entry: it lies at the intersection of 7 and 5 (SR and RS).
- Its old value is ∞ . Its new value is $7+5=12$.
- We select the smaller of the new and old values for Q_1 , i.e., 12.
- The resulting path is found by concatenating SR and RS, dropping one of the repeating R's: SRS.



EXECUTION OF SPA



$$Q_0 = \begin{matrix} & \begin{matrix} R & S & T & U \end{matrix} \\ \begin{matrix} R \\ S \\ T \\ U \end{matrix} & \begin{bmatrix} 7 & 5 & \infty & \infty \\ 7 & \infty & \infty & 2 \\ \infty & 3 & \infty & \infty \\ 4 & \infty & 1 & \infty \end{bmatrix} \end{matrix} \begin{bmatrix} RR & RS & - & - \\ SR & - & - & SU \\ - & TS & - & - \\ UR & - & UT & - \end{bmatrix}$$

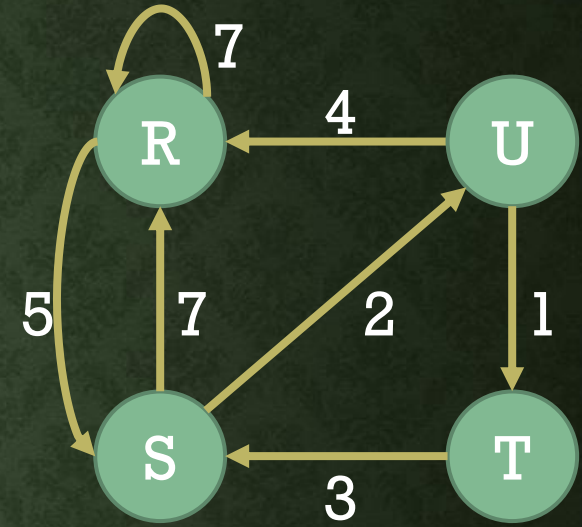
- Fix the first row and the first column to compute Q_1 .

$$Q_1 = \begin{matrix} & \begin{matrix} R & S & T & U \end{matrix} \\ \begin{matrix} R \\ S \\ T \\ U \end{matrix} & \begin{bmatrix} 7 & 5 & \infty & \infty \\ 7 & 12 & \blacksquare & \infty \\ \infty & & & \\ 4 & & & \end{bmatrix} \end{matrix} \begin{bmatrix} RR & RS & - & - \\ SR & SRS & \blacksquare & \\ - & & & \\ UR & & & \end{bmatrix}$$

- Concentrate on the ST entry: it lies at the intersection of 7 and ∞ (SR and -).
- Its old value is ∞ . Its new value is $7 + \infty = \infty$.
- We leave this entry unchanged.



EXECUTION OF SPA



$$Q_0 = \begin{matrix} & \begin{matrix} R & S & T & U \end{matrix} \\ \begin{matrix} R \\ S \\ T \\ U \end{matrix} & \begin{bmatrix} 7 & 5 & \infty & \infty \\ 7 & \infty & \infty & 2 \\ \infty & 3 & \infty & \infty \\ 4 & \infty & 1 & \infty \end{bmatrix} \end{matrix} \begin{bmatrix} RR & RS & - & - \\ SR & - & - & SU \\ - & TS & - & - \\ UR & - & UT & - \end{bmatrix}$$

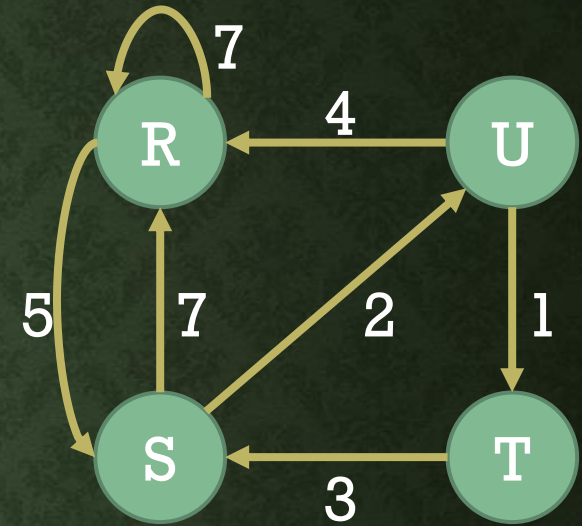
- Fix the first row and the first column to compute Q_1 .

$$Q_1 = \begin{matrix} & \begin{matrix} R & S & T & U \end{matrix} \\ \begin{matrix} R \\ S \\ T \\ U \end{matrix} & \begin{bmatrix} 7 & 5 & \infty & \infty \\ 7 & 12 & \infty & \blacksquare \\ \infty & & & \\ 4 & & & \end{bmatrix} \end{matrix} \begin{bmatrix} RR & RS & - & - \\ SR & SRS & \infty & \blacksquare \\ - & & & \\ UR & & & \end{bmatrix}$$

- Concentrate on the SU entry: it lies at the intersection of 7 and ∞ (SR and -).
- Its old value is 2. Its new value is $7 + \infty = \infty$.
- We select the smaller of the new and old values for Q_1 , i.e., 2.
- The old path is left as it is.



EXECUTION OF SPA



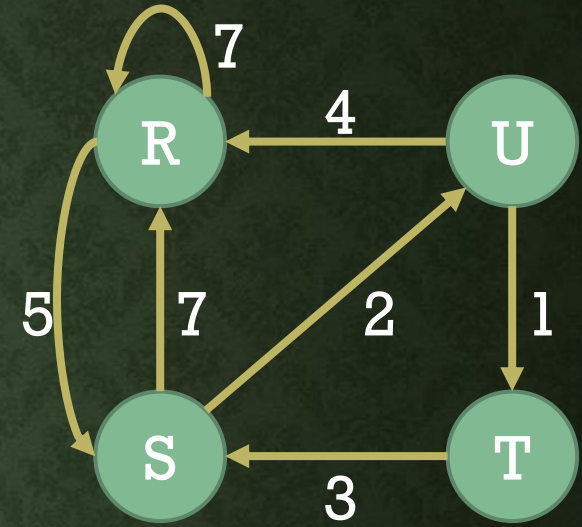
$$Q_0 = \begin{array}{c} \text{R} \\ \text{S} \\ \text{T} \\ \text{U} \end{array} \begin{array}{c} \text{R} \quad \text{S} \quad \text{T} \quad \text{U} \\ \begin{bmatrix} 7 & 5 & \infty & \infty \\ 7 & \infty & \infty & 2 \\ \infty & 3 & \infty & \infty \\ 4 & \infty & 1 & \infty \end{bmatrix} \end{array} \begin{array}{c} \text{RR} \quad \text{RS} \quad - \quad - \\ \text{SR} \quad - \quad - \quad \text{SU} \\ - \quad \text{TS} \quad - \quad - \\ \text{UR} \quad - \quad \text{UT} \quad - \end{array}$$

- Fix the first row and the first column to compute Q_1 .

$$Q_1 = \begin{array}{c} \text{R} \\ \text{S} \\ \text{T} \\ \text{U} \end{array} \begin{array}{c} \text{R} \quad \text{S} \quad \text{T} \quad \text{U} \\ \begin{bmatrix} 7 & 5 & \infty & \infty \\ 7 & 12 & \infty & 2 \\ \infty & 3 & \infty & \infty \\ 4 & 9 & 1 & \infty \end{bmatrix} \end{array} \begin{array}{c} \text{RR} \quad \text{RS} \quad - \quad - \\ \text{SR} \quad \text{SRS} \quad - \quad \text{SU} \\ - \quad \text{TS} \quad - \quad - \\ \text{UR} \quad \text{URS} \quad \text{UT} \quad - \end{array}$$



EXECUTION OF SPA



$$Q_1 = \begin{matrix} & \begin{matrix} R & S & T & U \end{matrix} \\ \begin{matrix} R \\ S \\ T \\ U \end{matrix} & \begin{bmatrix} 7 & 5 & \infty & \infty \\ 7 & 12 & \infty & 2 \\ \infty & 3 & \infty & \infty \\ 4 & 9 & 1 & \infty \end{bmatrix} \end{matrix} \begin{bmatrix} RR & RS & - & - \\ SR & SRS & - & SU \\ - & TS & - & - \\ UR & URS & UT & - \end{bmatrix}$$

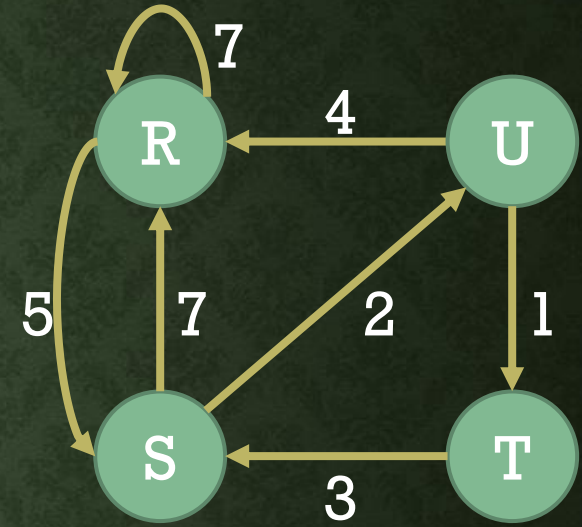
- Fix the second row and the second column to compute Q_2 .

$$Q_2 = \begin{matrix} & \begin{matrix} R & S & T & U \end{matrix} \\ \begin{matrix} R \\ S \\ T \\ U \end{matrix} & \begin{bmatrix} \boxed{7} & 5 & \infty & 2 \\ 7 & 12 & \infty & 2 \\ & 3 & & \\ & 9 & & \end{bmatrix} \end{matrix} \begin{bmatrix} \boxed{7} & RS & & \\ SR & SRS & - & SU \\ & TS & & \\ & URS & & \end{bmatrix}$$

- Concentrate on the RR entry: it lies at the intersection of 5 and 7 (RS and SR).
- Its old value is 7. Its new value is $5+7=12$.
- We select the smaller of the new and old values for Q_2 , i.e., 7.
- The old path is left as it is.



EXECUTION OF SPA



$$Q_1 = \begin{matrix} & \begin{matrix} R & S & T & U \end{matrix} \\ \begin{matrix} R \\ S \\ T \\ U \end{matrix} & \begin{bmatrix} 7 & 5 & \infty & \infty \\ 7 & 12 & \infty & 2 \\ \infty & 3 & \infty & \infty \\ 4 & 9 & 1 & \infty \end{bmatrix} \end{matrix} \begin{bmatrix} RR & RS & - & - \\ SR & SRS & - & SU \\ - & TS & - & - \\ UR & URS & UT & - \end{bmatrix}$$

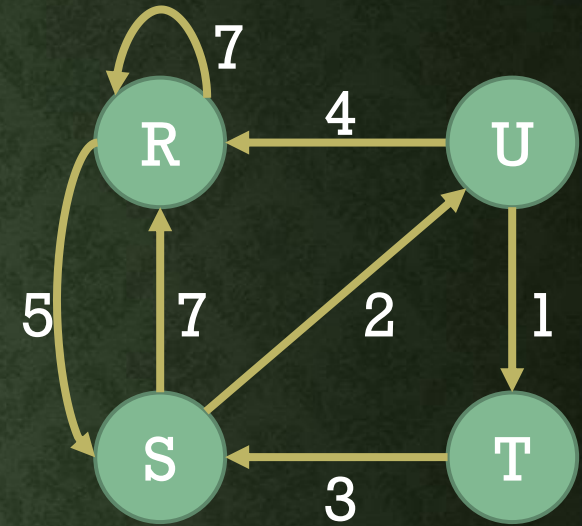
- Fix the second row and the second column to compute Q_2 .

$$Q_2 = \begin{matrix} & \begin{matrix} R & S & T & U \end{matrix} \\ \begin{matrix} R \\ S \\ T \\ U \end{matrix} & \begin{bmatrix} 7 & 5 & \infty & \boxed{7} \\ 7 & 12 & \infty & 2 \\ & 3 & & \\ & 9 & & \end{bmatrix} \end{matrix} \begin{bmatrix} RR & RS & - & \boxed{7} \\ SR & SRS & - & SU \\ & TS & & \\ & URS & & \end{bmatrix}$$

- Concentrate on the RU entry: it lies at the intersection of 5 and 2 (RS and SU).
- Its old value is ∞ . Its new value is $5+2=7$.
- We select the smaller of the new and old values for Q_2 , i.e., 7.
- The resulting path is found by concatenating RS and SU, dropping one of the repeating S's: RSU.



EXECUTION OF SPA



$$Q_1 = \begin{matrix} & \begin{matrix} R & S & T & U \end{matrix} \\ \begin{matrix} R \\ S \\ T \\ U \end{matrix} & \begin{bmatrix} 7 & 5 & \infty & \infty \\ 7 & 12 & \infty & 2 \\ \infty & 3 & \infty & \infty \\ 4 & 9 & 1 & \infty \end{bmatrix} \end{matrix} \begin{bmatrix} RR & RS & - & - \\ SR & SRS & - & SU \\ - & TS & - & - \\ UR & URS & UT & - \end{bmatrix}$$

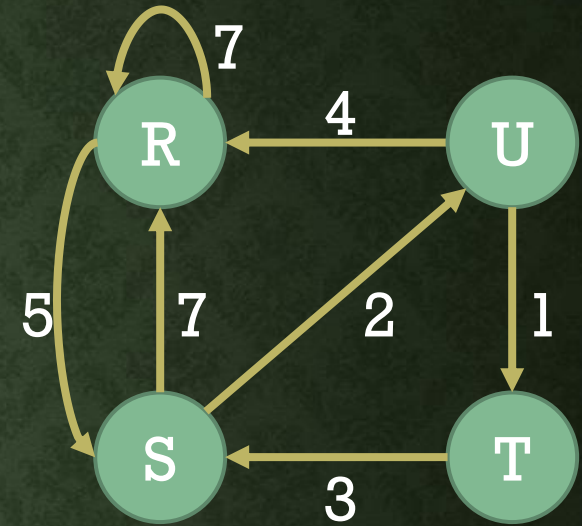
- Fix the second row and the second column to compute Q_2 .

$$Q_2 = \begin{matrix} & \begin{matrix} R & S & T & U \end{matrix} \\ \begin{matrix} R \\ S \\ T \\ U \end{matrix} & \begin{bmatrix} 7 & 5 & \infty & 7 \\ 7 & 12 & \infty & 2 \\ \blacksquare & 3 & & \\ \blacksquare & 9 & & \end{bmatrix} \end{matrix} \begin{bmatrix} RR & RS & - & RSU \\ SR & SRS & - & SU \\ \blacksquare & TS & & \\ \blacksquare & URS & & \end{bmatrix}$$

- Concentrate on the TR entry: it lies at the intersection of 3 and 7 (TS and SR).
- Its old value is ∞ . Its new value is $3+7=10$.
- We select the smaller of the new and old values for Q_2 , i.e., 10.
- The resulting path is found by concatenating TS and SR, dropping one of the repeating S's: TSR.



EXECUTION OF SPA



$$Q_1 = \begin{matrix} & \begin{matrix} R & S & T & U \end{matrix} \\ \begin{matrix} R \\ S \\ T \\ U \end{matrix} & \begin{bmatrix} 7 & 5 & \infty & \infty \\ 7 & 12 & \infty & 2 \\ \infty & 3 & \infty & \infty \\ 4 & 9 & 1 & \infty \end{bmatrix} \end{matrix} \begin{bmatrix} RR & RS & - & - \\ SR & SRS & - & SU \\ - & TS & - & - \\ UR & URS & UT & - \end{bmatrix}$$

- Fix the second row and the second column to compute Q_2 .

$$Q_2 = \begin{matrix} & \begin{matrix} R & S & T & U \end{matrix} \\ \begin{matrix} R \\ S \\ T \\ U \end{matrix} & \begin{bmatrix} 7 & 5 & \infty & 7 \\ 7 & 12 & \infty & 2 \\ 10 & 3 & \infty & 5 \\ 4 & 9 & 1 & 11 \end{bmatrix} \end{matrix} \begin{bmatrix} RR & RS & - & RSU \\ SR & SRS & - & SU \\ TSR & TS & - & TSU \\ UR & URS & UT & URSU \end{bmatrix}$$

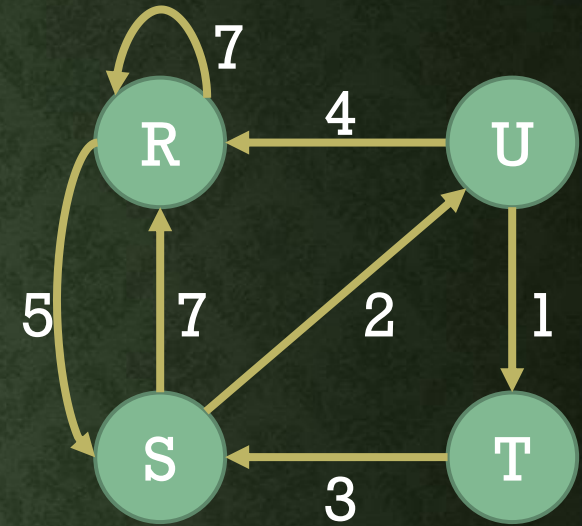


EXECUTION OF SPA

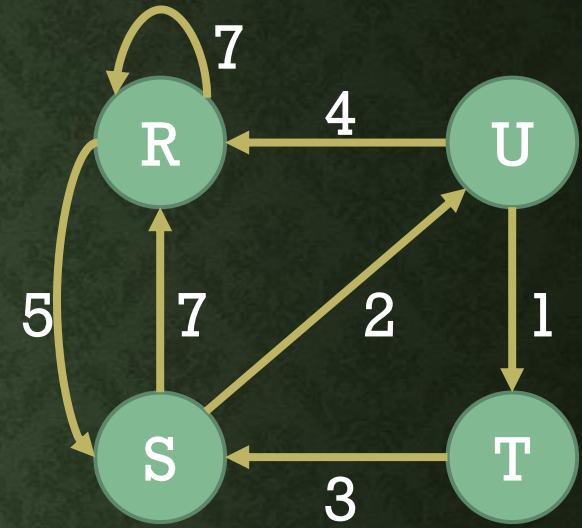
$$Q_2 = \begin{array}{c} \text{R} \\ \text{S} \\ \text{T} \\ \text{U} \end{array} \begin{array}{c} \text{R} \quad \text{S} \quad \text{T} \quad \text{U} \\ \left[\begin{array}{cccc} 7 & 5 & \infty & 7 \\ 7 & 12 & \infty & 2 \\ 10 & 3 & \infty & 5 \\ 4 & 9 & 1 & 11 \end{array} \right] \end{array} \begin{array}{c} \text{RR} \quad \text{RS} \quad - \quad \text{RSU} \\ \text{SR} \quad \text{SRS} \quad - \quad \text{SU} \\ \text{TSR} \quad \text{TS} \quad - \quad \text{TSU} \\ \text{UR} \quad \text{URS} \quad \text{UT} \quad \text{URSU} \end{array}$$

- Fix the third row and the third column to compute Q_3 .

$$Q_3 = \begin{array}{c} \text{R} \\ \text{S} \\ \text{T} \\ \text{U} \end{array} \begin{array}{c} \text{R} \quad \text{S} \quad \text{T} \quad \text{U} \\ \left[\begin{array}{cccc} & & \infty & \\ & & \infty & \\ 10 & 3 & \infty & 5 \\ & & 1 & \end{array} \right] \end{array} \begin{array}{c} - \\ - \\ \text{TSR} \quad \text{TS} \quad - \quad \text{TSU} \\ \text{UT} \end{array}$$



EXECUTION OF SPA



$$Q_2 = \begin{matrix} & \begin{matrix} R & S & T & U \end{matrix} \\ \begin{matrix} R \\ S \\ T \\ U \end{matrix} & \begin{bmatrix} 7 & 5 & \infty & 7 \\ 7 & 12 & \infty & 2 \\ 10 & 3 & \infty & 5 \\ 4 & 9 & 1 & 11 \end{bmatrix} \end{matrix} \begin{bmatrix} RR & RS & - & RSU \\ SR & SRS & - & SU \\ TSR & TS & - & TSU \\ UR & URS & UT & URSU \end{bmatrix}$$

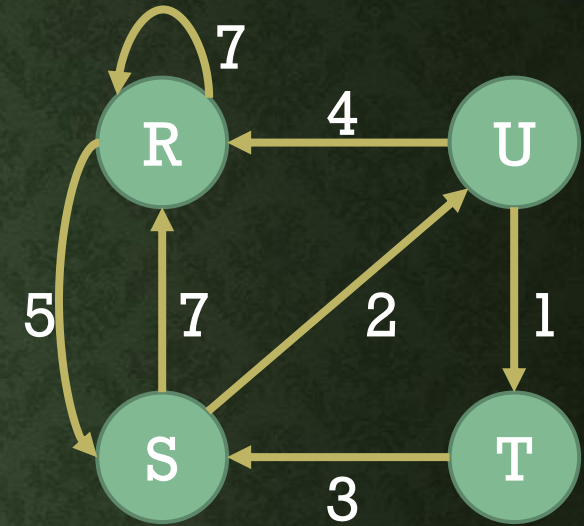
- Fix the third row and the third column to compute Q_3 .

$$Q_3 = \begin{matrix} & \begin{matrix} R & S & T & U \end{matrix} \\ \begin{matrix} R \\ S \\ T \\ U \end{matrix} & \begin{bmatrix} 7 & 5 & \infty & 7 \\ 7 & 12 & \infty & 2 \\ 10 & 3 & \infty & 5 \\ \blacksquare & & 1 & \blacksquare \end{bmatrix} \end{matrix} \begin{bmatrix} RR & RS & - & RSU \\ SR & SRS & - & SU \\ \text{TSR} & \text{TS} & - & \text{TSU} \\ \blacksquare & & \text{UT} & \end{bmatrix}$$

- Concentrate on the UR entry: it lies at the intersection of 1 and 10 (UT and TSR).
- Its old value is 4. Its new value is $1+10=11$.
- We select the smaller of the new and old values for Q_3 , i.e., 4.
- The old path is left as it is.



EXECUTION OF SPA



$$Q_2 = \begin{matrix} & \begin{matrix} R & S & T & U \end{matrix} \\ \begin{matrix} R \\ S \\ T \\ U \end{matrix} & \begin{bmatrix} 7 & 5 & \infty & 7 \\ 7 & 12 & \infty & 2 \\ 10 & 3 & \infty & 5 \\ 4 & 9 & 1 & 11 \end{bmatrix} \end{matrix} \begin{bmatrix} RR & RS & - & RSU \\ SR & SRS & - & SU \\ TSR & TS & - & TSU \\ UR & URS & UT & URSU \end{bmatrix}$$

- Fix the third row and the third column to compute Q_3 .

$$Q_3 = \begin{matrix} & \begin{matrix} R & S & T & U \end{matrix} \\ \begin{matrix} R \\ S \\ T \\ U \end{matrix} & \begin{bmatrix} 7 & 5 & \infty & 7 \\ 7 & 12 & \infty & 2 \\ 10 & 3 & \infty & 5 \\ 4 & \blacksquare & 1 & \blacksquare \end{bmatrix} \end{matrix} \begin{bmatrix} RR & RS & - & RSU \\ SR & SRS & - & SU \\ \textcolor{red}{TSR} & \textcolor{red}{TS} & - & \textcolor{red}{TSU} \\ UR & \blacksquare & \textcolor{red}{UT} & \end{bmatrix}$$

- Concentrate on the US entry: it lies at the intersection of 1 and 3 (UT and TS).
- Its old value is 9. Its new value is $1+3=4$.
- We select the smaller of the new and old values for Q_3 , i.e., 4.
- The resulting path is found by concatenating UT and TS, dropping one of the repeating T's: UTS.

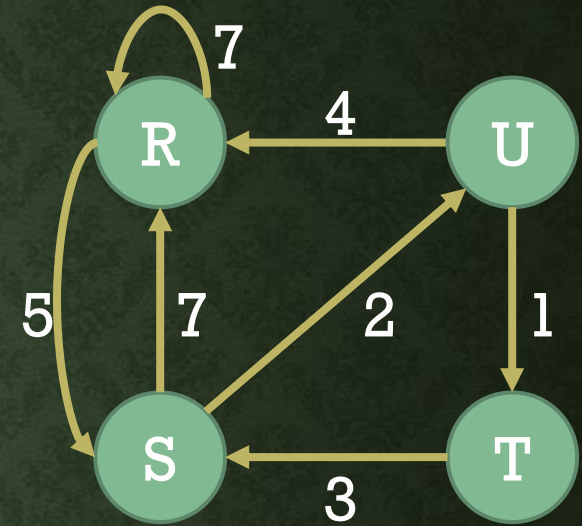


EXECUTION OF SPA

$$Q_2 = \begin{matrix} & \begin{matrix} R & S & T & U \end{matrix} \\ \begin{matrix} R \\ S \\ T \\ U \end{matrix} & \begin{bmatrix} 7 & 5 & \infty & 7 \\ 7 & 12 & \infty & 2 \\ 10 & 3 & \infty & 5 \\ 4 & 9 & 1 & 11 \end{bmatrix} \end{matrix} \begin{bmatrix} RR & RS & - & RSU \\ SR & SRS & - & SU \\ TSR & TS & - & TSU \\ UR & URS & UT & URSU \end{bmatrix}$$

- Fix the third row and the third column to compute Q_3 .

$$Q_3 = \begin{matrix} & \begin{matrix} R & S & T & U \end{matrix} \\ \begin{matrix} R \\ S \\ T \\ U \end{matrix} & \begin{bmatrix} 7 & 5 & \infty & 7 \\ 7 & 12 & \infty & 2 \\ 10 & 3 & \infty & 5 \\ 4 & 4 & 1 & 6 \end{bmatrix} \end{matrix} \begin{bmatrix} RR & RS & - & RSU \\ SR & SRS & - & SU \\ \text{TSR} & \text{TS} & - & \text{TSU} \\ UR & UTS & \text{UT} & UTSU \end{bmatrix}$$

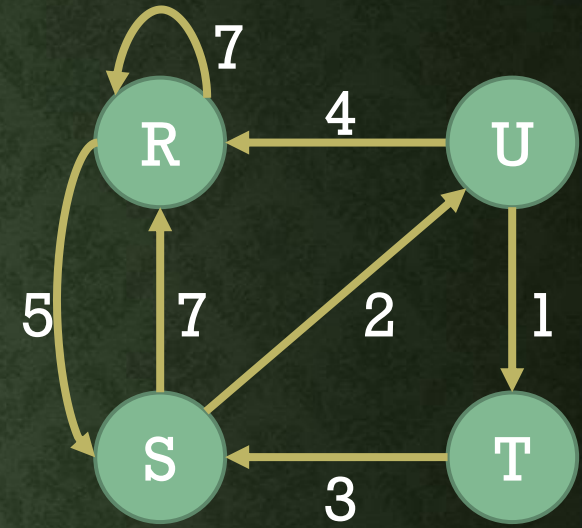


EXECUTION OF SPA

$$Q_3 = \begin{matrix} & \begin{matrix} R & S & T & U \end{matrix} \\ \begin{matrix} R \\ S \\ T \\ U \end{matrix} & \begin{bmatrix} 7 & 5 & \infty & 7 \\ 7 & 12 & \infty & 2 \\ 10 & 3 & \infty & 5 \\ 4 & 4 & 1 & 6 \end{bmatrix} \end{matrix} \begin{bmatrix} RR & RS & - & RSU \\ SR & SRS & - & SU \\ TSR & TS & - & TSU \\ UR & UTS & UT & UTSU \end{bmatrix}$$

- Fix the fourth row and the fourth column to compute Q_4 .

$$Q_4 = \begin{matrix} & \begin{matrix} R & S & T & U \end{matrix} \\ \begin{matrix} R \\ S \\ T \\ U \end{matrix} & \begin{bmatrix} & & & 7 \\ & & & 2 \\ & & & 5 \\ 4 & 4 & 1 & 6 \end{bmatrix} \end{matrix} \begin{bmatrix} & & & RSU \\ & & & SU \\ & & & TSU \\ UR & UTS & UT & UTSU \end{bmatrix}$$

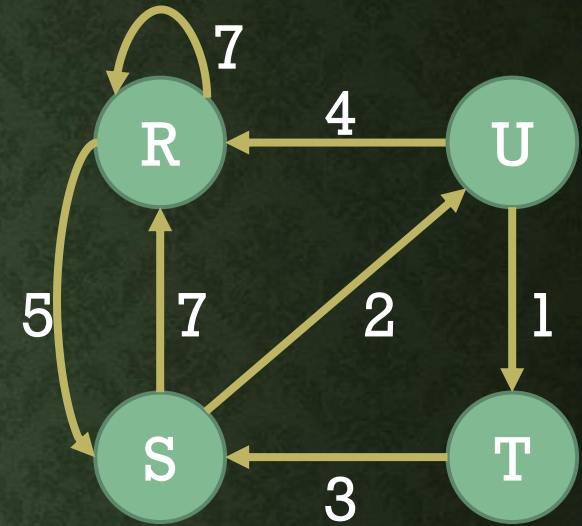


EXECUTION OF SPA

$$Q_3 = \begin{matrix} & \begin{matrix} R & S & T & U \end{matrix} \\ \begin{matrix} R \\ S \\ T \\ U \end{matrix} & \begin{bmatrix} 7 & 5 & \infty & 7 \\ 7 & 12 & \infty & 2 \\ 10 & 3 & \infty & 5 \\ 4 & 4 & 1 & 6 \end{bmatrix} \end{matrix} \begin{bmatrix} RR & RS & - & RSU \\ SR & SRS & - & SU \\ TSR & TS & - & TSU \\ UR & UTS & UT & UTSU \end{bmatrix}$$

- Fix the fourth row and the fourth column to compute Q_4 .

$$Q_4 = \begin{matrix} & \begin{matrix} R & S & T & U \end{matrix} \\ \begin{matrix} R \\ S \\ T \\ U \end{matrix} & \begin{bmatrix} 7 & 5 & 8 & 7 \\ & & & 2 \\ & & & 5 \\ 4 & 4 & 1 & 6 \end{bmatrix} \end{matrix} \begin{bmatrix} RR & RS & RSUT & RSU \\ & & & SU \\ & & & TSU \\ UR & UTS & UT & UTSU \end{bmatrix}$$

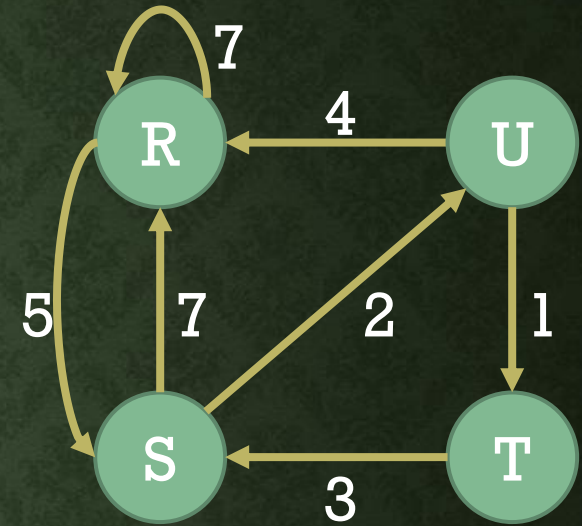


EXECUTION OF SPA

$$Q_3 = \begin{array}{c} \text{R} \\ \text{S} \\ \text{T} \\ \text{U} \end{array} \begin{array}{c} \text{R} \quad \text{S} \quad \text{T} \quad \text{U} \\ \left[\begin{array}{cccc} 7 & 5 & \infty & 7 \\ 7 & 12 & \infty & 2 \\ 10 & 3 & \infty & 5 \\ 4 & 4 & 1 & 6 \end{array} \right] \end{array} \begin{array}{c} \text{RR} \quad \text{RS} \quad - \quad \text{RSU} \\ \text{SR} \quad \text{SRS} \quad - \quad \text{SU} \\ \text{TSR} \quad \text{TS} \quad - \quad \text{TSU} \\ \text{UR} \quad \text{UTS} \quad \text{UT} \quad \text{UTSU} \end{array}$$

- Fix the fourth row and the fourth column to compute Q_4 .

$$Q_4 = \begin{array}{c} \text{R} \\ \text{S} \\ \text{T} \\ \text{U} \end{array} \begin{array}{c} \text{R} \quad \text{S} \quad \text{T} \quad \text{U} \\ \left[\begin{array}{cccc} 7 & 5 & 8 & 7 \\ 6 & 6 & 3 & 2 \\ & & & 5 \\ 4 & 4 & 1 & 6 \end{array} \right] \end{array} \begin{array}{c} \text{RR} \quad \text{RS} \quad \text{RSUT} \quad \text{RSU} \\ \text{SUR} \quad \text{SUTS} \quad \text{SUT} \quad \text{SU} \\ & & & \text{TSU} \\ \text{UR} \quad \text{UTS} \quad \text{UT} \quad \text{UTSU} \end{array}$$



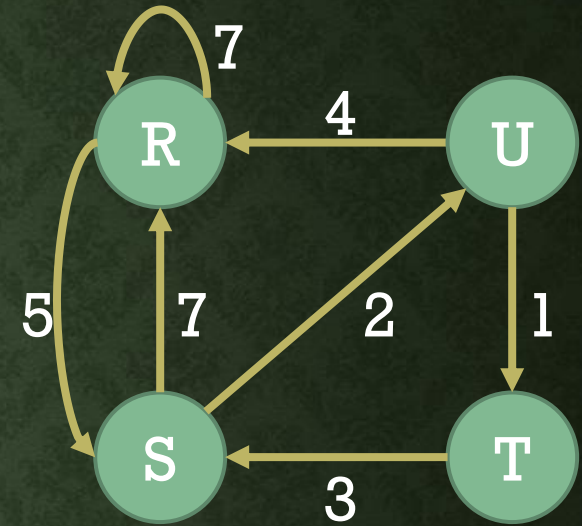
EXECUTION OF SPA

$$Q_3 = \begin{matrix} & \begin{matrix} R & S & T & U \end{matrix} \\ \begin{matrix} R \\ S \\ T \\ U \end{matrix} & \begin{bmatrix} 7 & 5 & \infty & 7 \\ 7 & 12 & \infty & 2 \\ 10 & 3 & \infty & 5 \\ 4 & 4 & 1 & 6 \end{bmatrix} \end{matrix} \begin{bmatrix} RR & RS & - & RSU \\ SR & SRS & - & SU \\ TSR & TS & - & TSU \\ UR & UTS & UT & UTSU \end{bmatrix}$$

- Fix the fourth row and the fourth column to compute Q_4 .

$$Q_4 = \begin{matrix} & \begin{matrix} R & S & T & U \end{matrix} \\ \begin{matrix} R \\ S \\ T \\ U \end{matrix} & \begin{bmatrix} 7 & 5 & 8 & 7 \\ 6 & 6 & 3 & 2 \\ 9 & 3 & 6 & 5 \\ 4 & 4 & 1 & 6 \end{bmatrix} \end{matrix} \begin{bmatrix} RR & RS & RSUT & RSU \\ SUR & SUTS & SUT & SU \\ TSUR & TS & TSUT & TSU \\ UR & UTS & UT & UTSU \end{bmatrix}$$

- This matrix shows the shortest paths between all node pairs.



ANALYSIS OF SPA

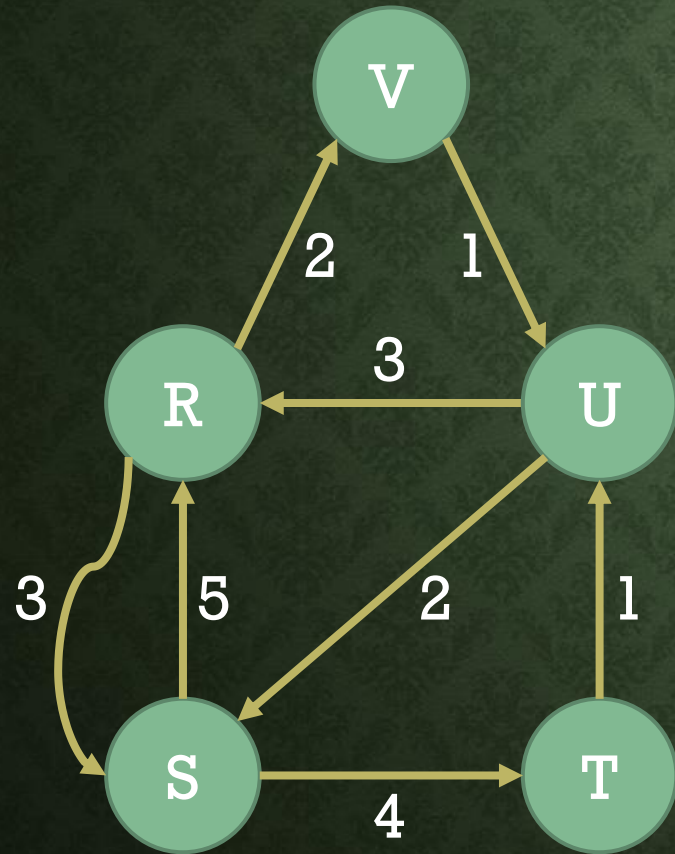
- There are n iterations in SPA.
- SPA computes an $n \times n$ matrix in each iteration, where n is the number of nodes.
- In each iteration, it uses 2 times the space taken by a matrix of $n \times n$.

Time: $O(n^3)$

Space: $O(n^2)$



HOMEWORK



For the weighted digraph shown here, find the weight matrix.

Execute the shortest path algorithm for the given weighted digraph.



WHAT DID WE LEARN TODAY?

We found the weight matrix
for a weighted digraph.

We executed the
SPA on a weighted
digraph.

We investigated
the time and
space
complexities of
SPA.



THINGS TO DO



Read the book!



Note your questions and put them up in the relevant online session.



Email suggestions on content or quality of this lecture at uroojain@neduet.edu.pk

